Google Cloud

Workspace Transformation Summary

Course

Google Workspace – Work Transformation

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Course summary

01

Day 1

- Methodology
- Provisioning
- System Access & Authentication
- Mail Routing

02

Day 2

- Migration
- Coexistence
- Security

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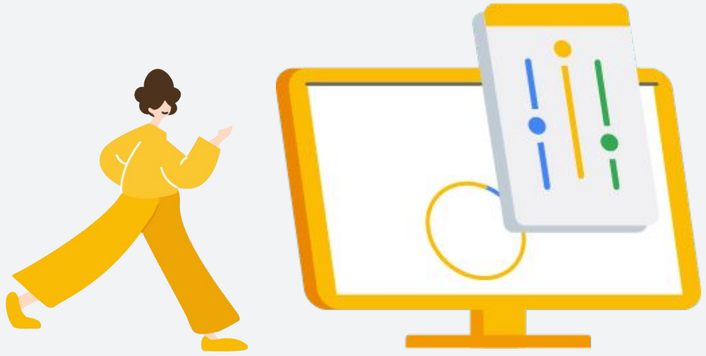
Day 1



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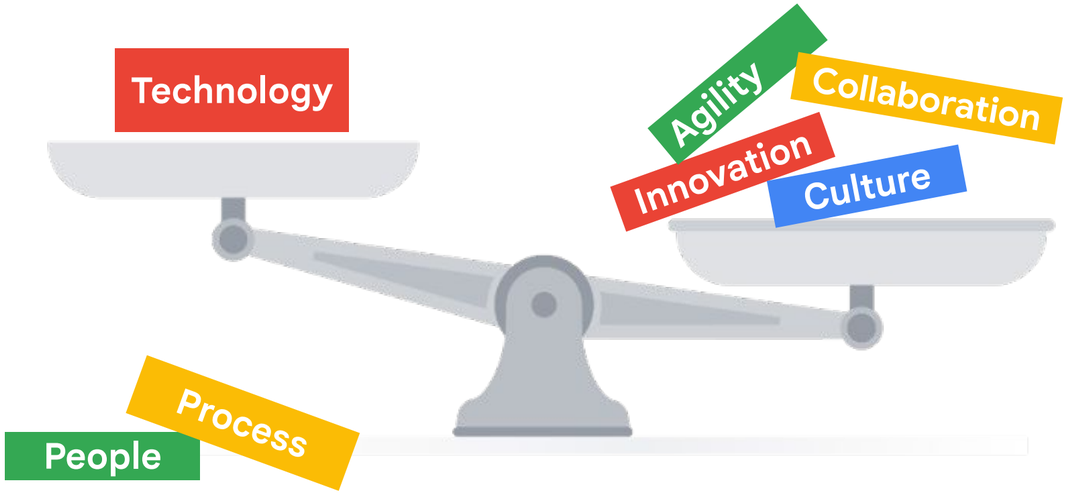
Fail #1:

The project is perceived as being about technology

- Getting technology right is crucial, but focusing solely on technical aspects can hinder success.
- Successful projects dedicate over 60% of time on people, process, and cultural aspects.
- Work transformation cannot be achieved solely by replacing technology.



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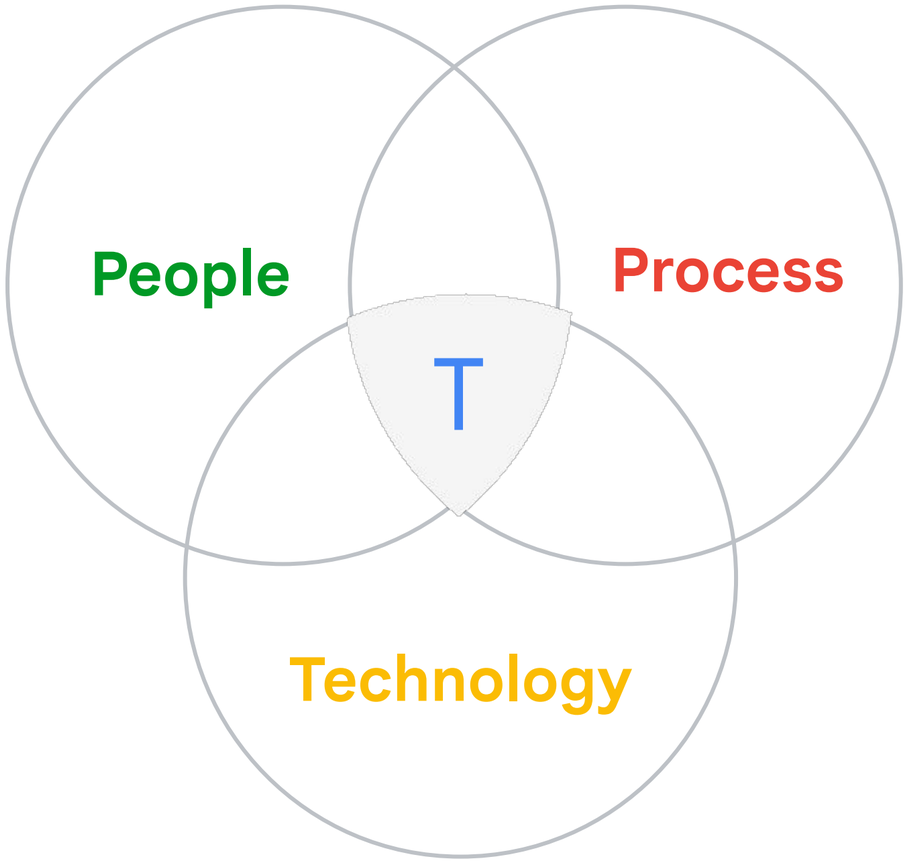
Every deployment has three dimensions

Work **transformation** takes place when people, process and technology align.



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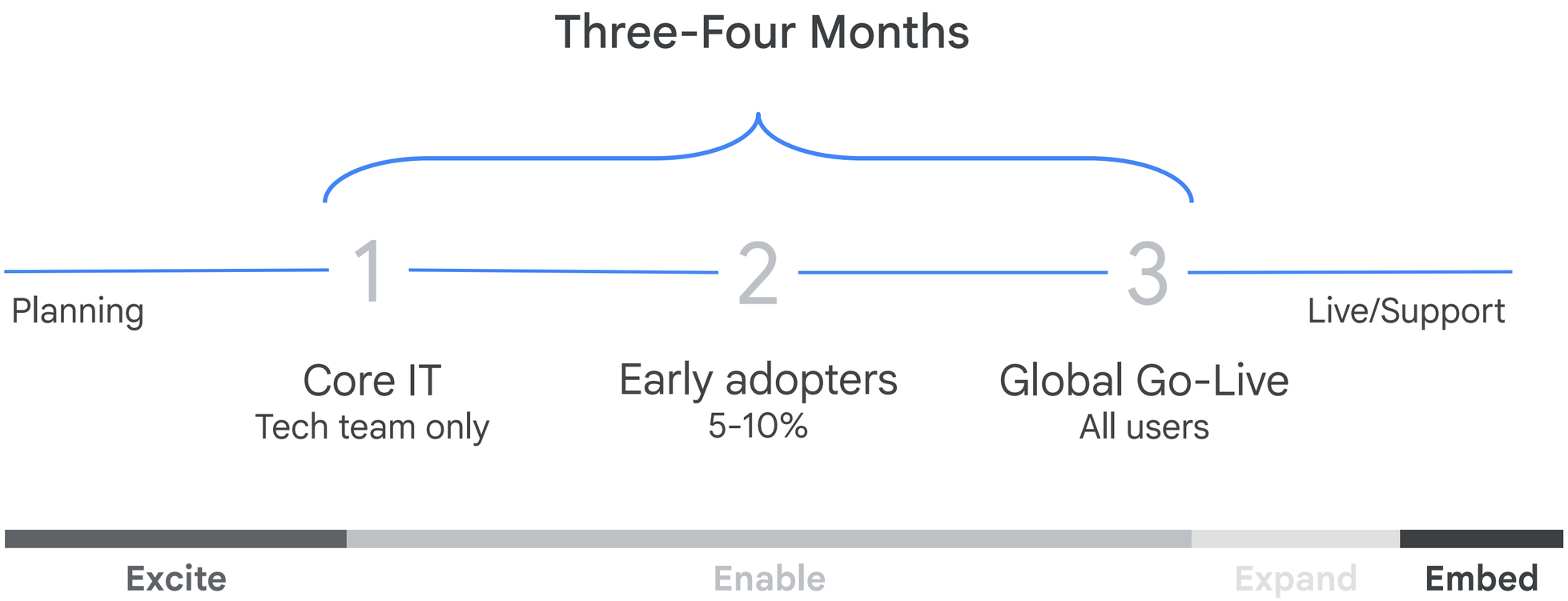
Path to a successful Google Workspace Deployment

3-phase migration model



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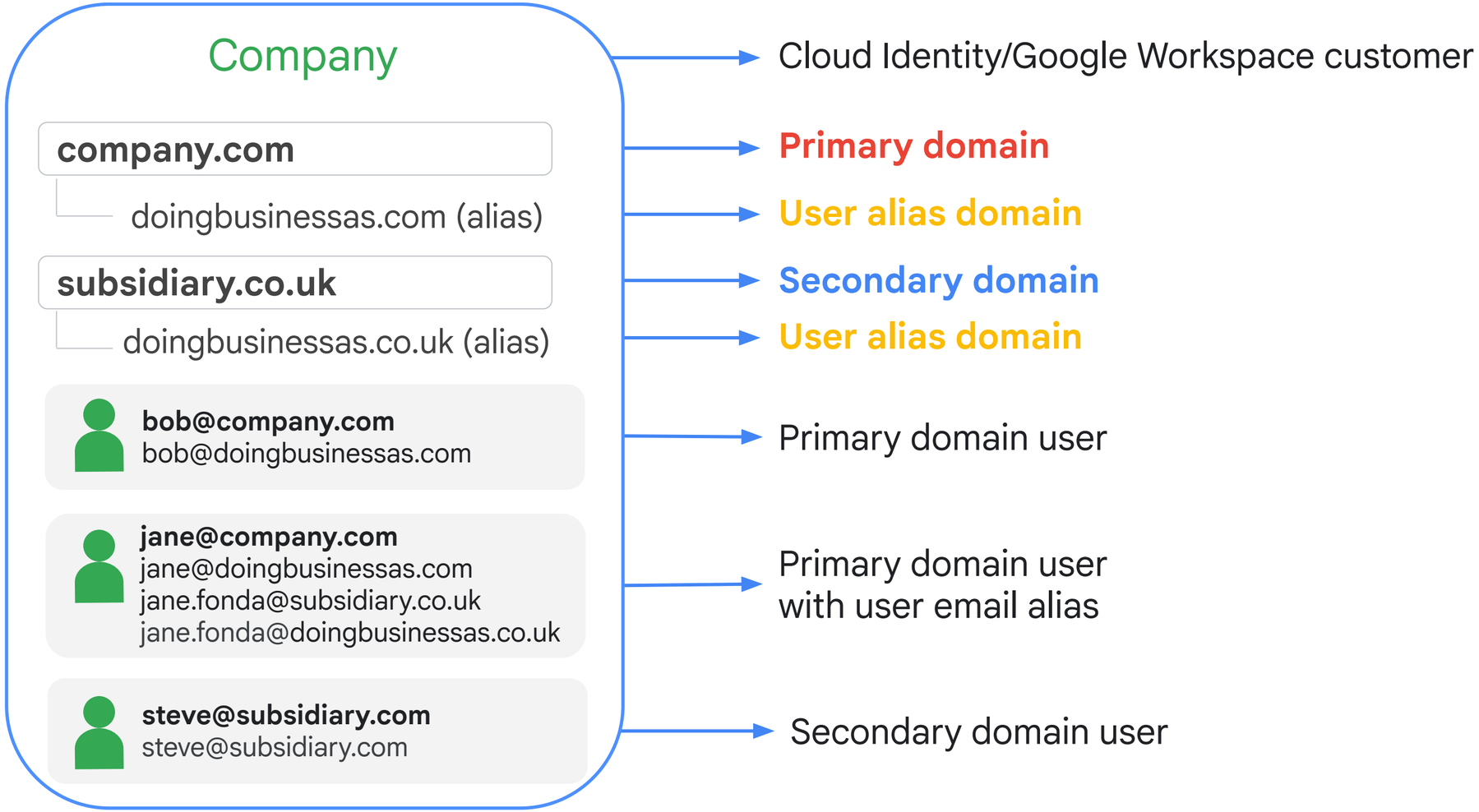
Domain provision key terminology



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Security is increasingly difficult

It's becoming increasingly difficult for organizations to protect assets at scale.

Phishing attacks

Targeted threats are very tough to detect.

Keeping up

Attackers rapidly change tactics to defeat email security measures.

Lack of control

Lack of admin controls on user emails makes remediation hard.

Data exfiltration

Accidental or intentional sharing of corporate info via email.

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User provisioning options

Method	Effort	Staff	Notes
Manual provisioning	High	Google Workspace Admin	Easiest method, but not scalable
CSV upload via Admin Console	Medium	Google Workspace Admin	More flexibility, but not scalable
Google Cloud Directory Sync	**Medium**	**LDAP Admin**	**Integrates with LDAP, scalable, requires no programming**
Third-party tools (Okta, Ping, Azure, etc.)	Medium	LDAP Admin	Scalable, may incur additional cost
Admin SDK Directory API	High	LDAP Admin Dev Staff	Scalable, flexible, requires in-depth programming

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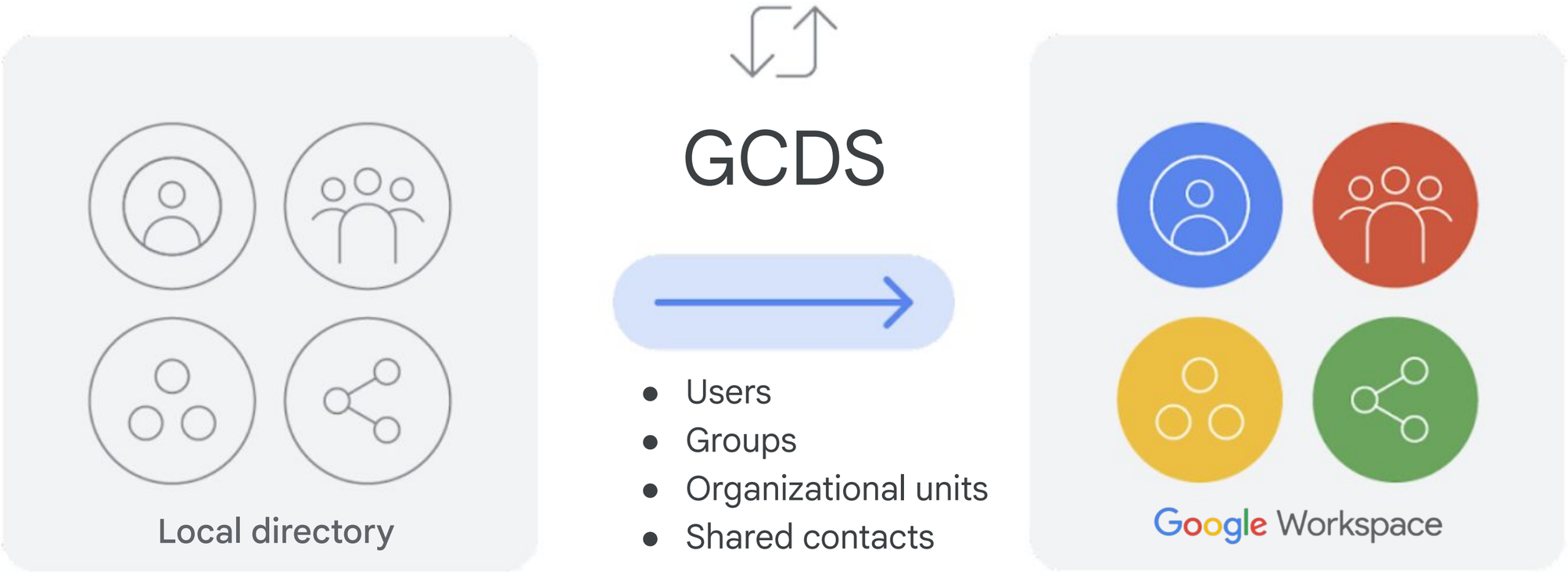
GCDS

- GCDS supports any Lightweight Directory Access Protocol (LDAP) directory server
- It performs a one-way synchronization, ensuring that Google domain data matches the local directory.



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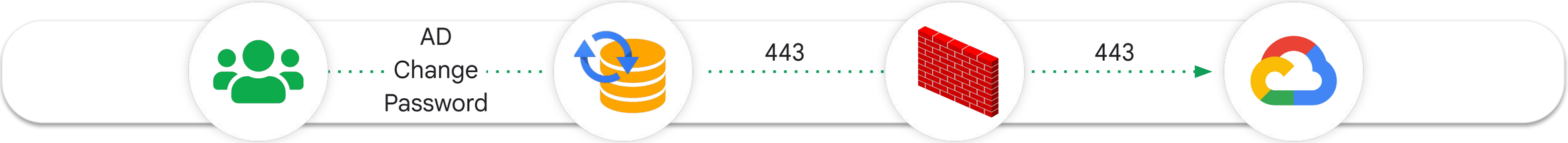
Password Sync

User provisioning



Overview

- Synchronizes user password from MS Active Directory to Google Workspace as they're changed – in real time
  - User changes password
  - Password Sync intercepts the raw password and applies a salted SHA-512 hash
  - Encrypted via TLS, salted hash is sent to Google using the Directory API

Benefits

- Passwords sync in real time - securely using native APIs
- Mitigates account lock-out if third-party IDP is down

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User authentication scenarios

Authentication

Google authentication

01

- Recommended 2SV (mandatory for administrators with access to Admin Console)

SSO with Third-Party Identity Provider

02

- Google as a SAML Service Provider | Recommended 2SV from Google or third party

Google as an SSO Identity Provider

03

- Recommended 2SV from Google

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2-step verification

Authentication

Google recommends security keys for best in class security

SMS / Voice	Backup codes	Authenticator (TOTP)	Google prompt (Mobile Push)	Security Key
Phishable SS7 vuln SIM Swap	Phishable			Phishing-resistant

Increased Security

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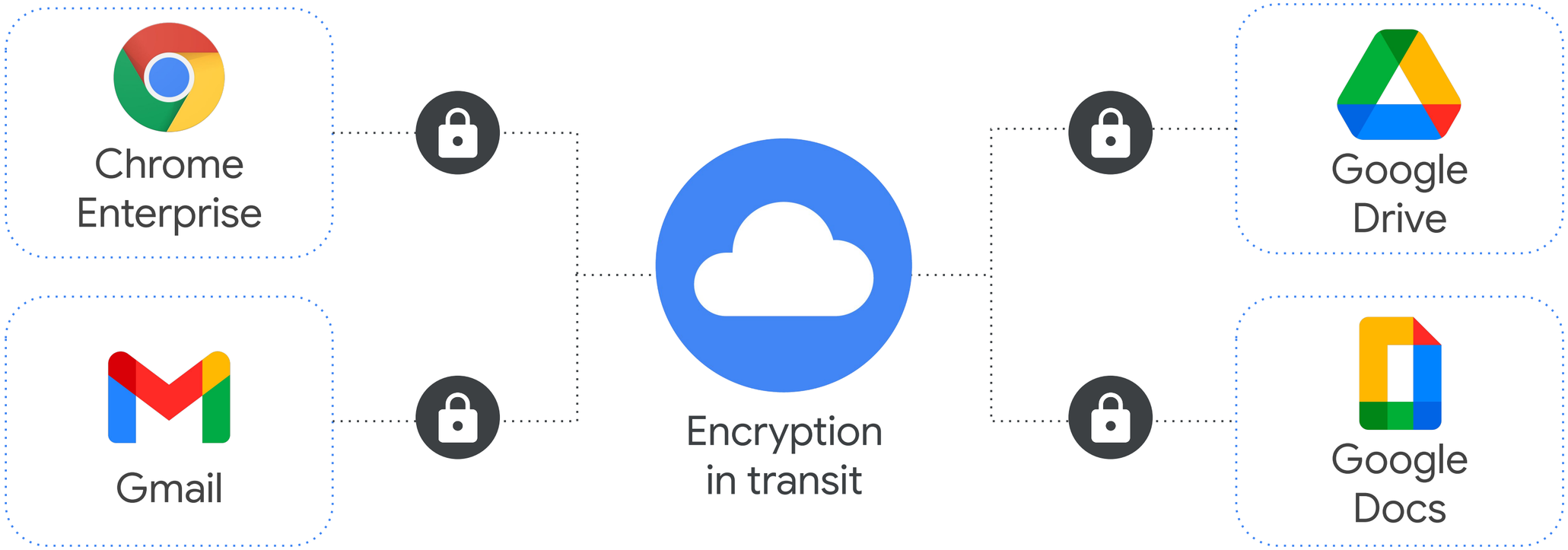
Cloud solution

A pure-based cloud solution can help mitigate risks and reduce cost.



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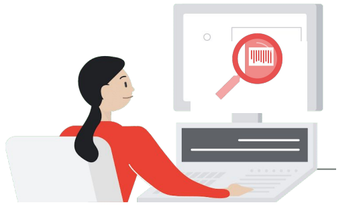
Administration accounts

Administration

Super administrator

- Require 2SV for Super Admin accounts
  * Google recommends security keys
- Apply the principle of least privilege and use delegate admins where possible
- Do not use admin accounts for daily work
- Limit the number of Super Admin accounts

Delegated administrator



- Enforce 2-step on delegated admins in-line with super administrator policies
- Assign administrative roles to different support tiers or geographies
- Create custom roles that suit your organizations custom admin needs

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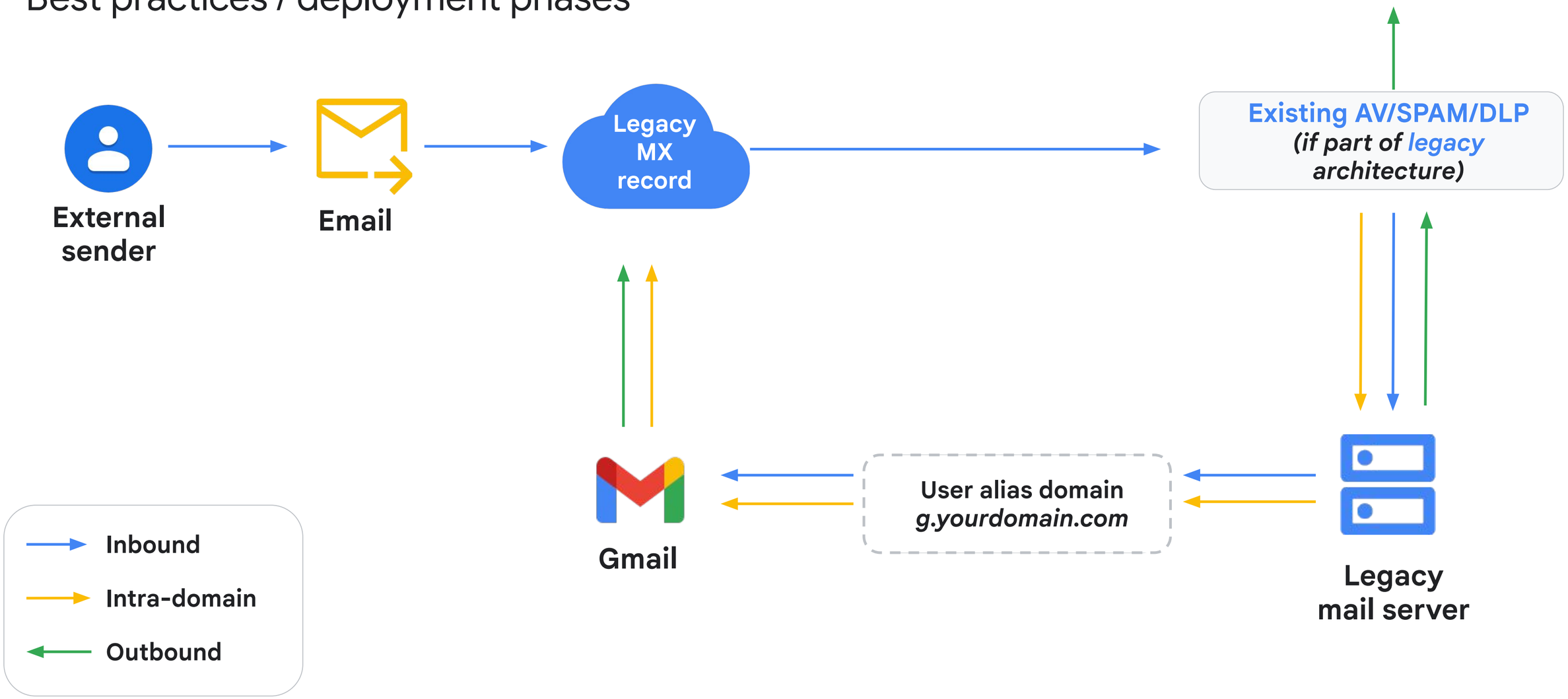
Core IT (dual delivery)

Best practices / deployment phases



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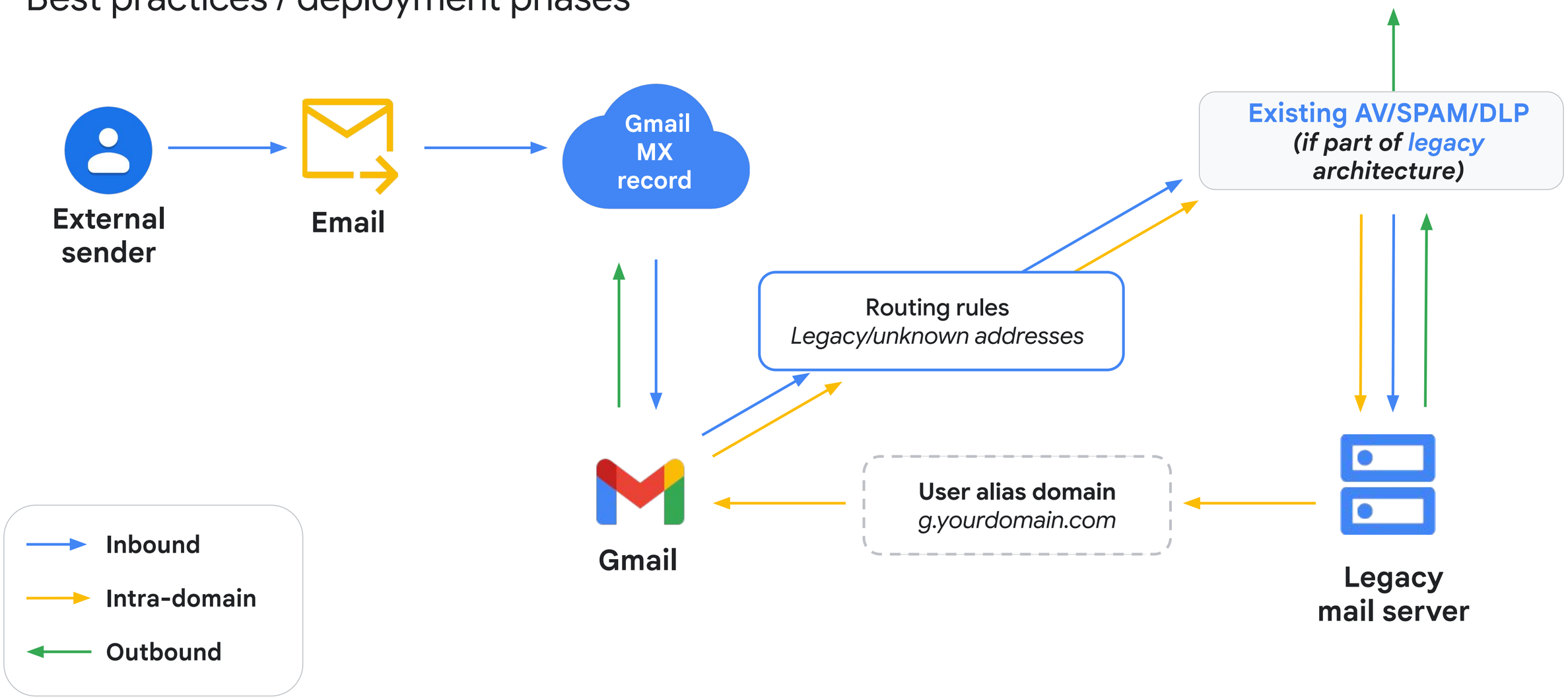
Early adopters (split delivery)

Best practices / deployment phases



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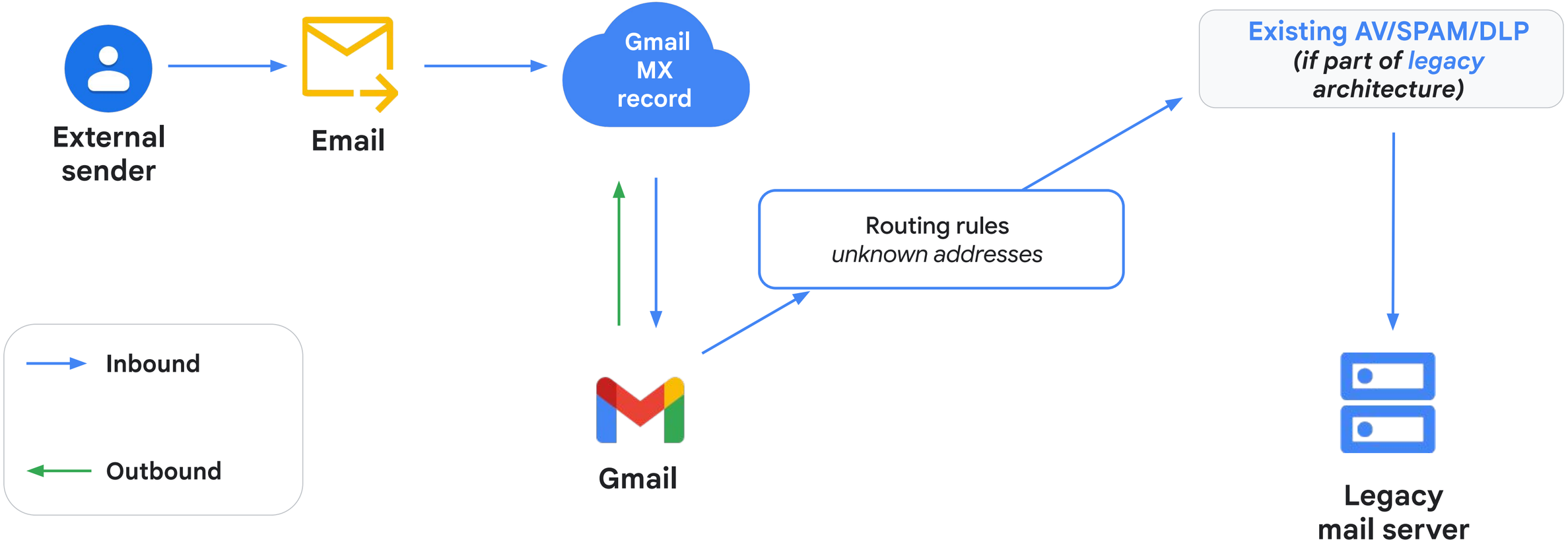
Global go-live (direct delivery)

Best practices / deployment phases



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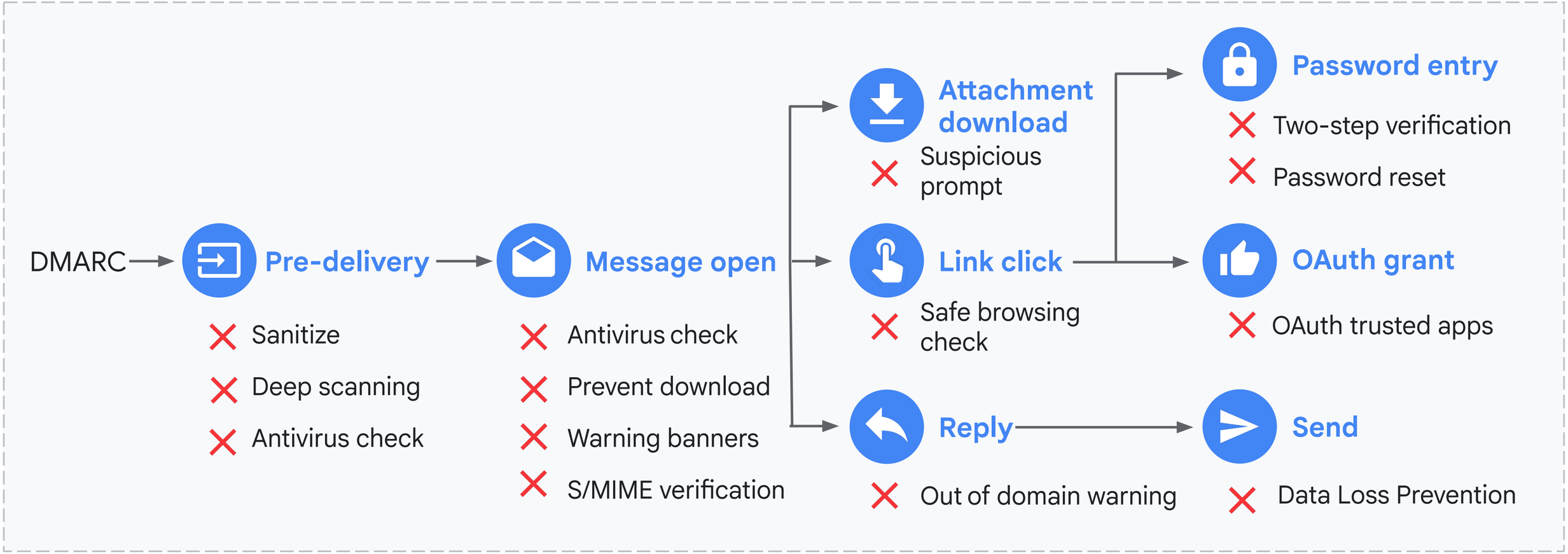
Message security flow

Email security

99.9%+ accuracy in blocking spam and business email compromise scams



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SPF, DKIM, and DMARC summary

SPF

- It will advertise which IPs are allowed to send message on your domain's behalf.

DKIM

- It will confirm that the content of the message is trustworthy and will confirm that it has not be changed after sending the message.

DMARC

- It will direct what should be done to the received message based on the published SPF and DKIM and will enforce the policies you have set.

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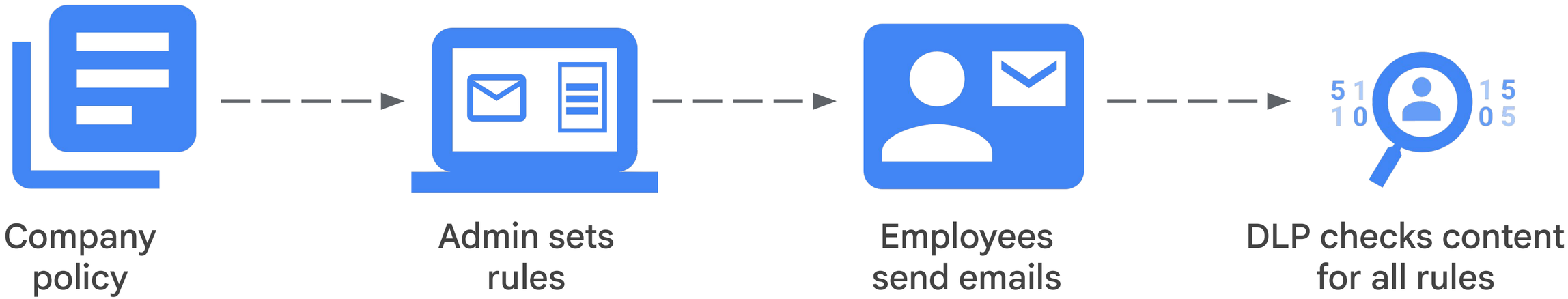
Data Loss Prevention (DLP) tools



DLP tools check emails comply with the rules

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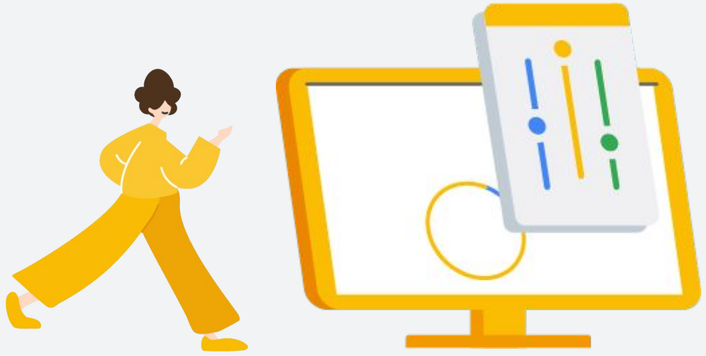
Day 2



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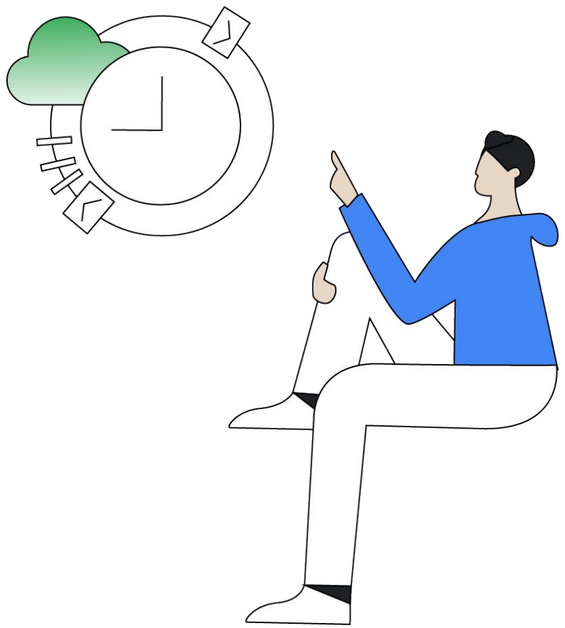
Migrate as little as possible for Go-Live

- Prioritize Day 1 productivity in data migration.
- Identify essential data for a smooth Go-Live.
- Migrate only necessary data to minimize disruption.
- Consider leaving non-essential data migration post Go-Live.



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Migrate as little as possible for Go-Live

Option 1

- Cost-conscious customers often prefer Greenfield (minimal migration), which offers a fresh start and potential cost saving.

Option 2

- Risk-averse customers may opt for partial migration, but it involves higher costs and complexities.

Option 3

- Risk-averse customers may choose complete migration to ensure access to historical data, but it involves higher costs and complexities.

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Migrate as little as possible for Go-Live

Before Go-Live				Post Go-Live
Option 1: Greenfield	**Option 2: Partial**	**Option 3: Complete**		
Green field ● No data migration Calendar and contacts ● Migrates quickly ● Can be admin or user initiated	Mail ● 30 days ● Up to one year Calendar ● All future events Contacts ● All contacts	Complete migration ● Not recommended ● Costly ● Extends timelines ● Resource intensive	**Go Live**	Additional mail history ● Mailbox backfill (during off hours) ● Personal archives ○ .pst files ○ .nsf files

Recommendation: reduce churn for aggressive go-live timelines by limiting migration, and giving users read-only access to legacy systems.

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Migration tools overview: admin led migrations

	Google Workspace Migrate (GWM)	Google Workspace Migration for Microsoft Exchange (GWMME)	Data Migration Service (DMS)
Key characteristics	• Customer installed and managed • Scalable to enterprise projects	• Customer installed and managed • Scalable to medium size projects	• Set it up and control everything right in the Google Admin console
Who has access?	• Google Workspace Business SKU • Google Workspace Enterprise SKU • Google Workspace Essentials SKU	• All customers	• All customers
When to use?	• Complex / large migrations from Exchange / Microsoft 365, file migrations	• Microsoft Exchange (up to 2016) • Migration from IMAP or PST sources	• Typically for smaller migration (less than 100 users) and a simple, self-guided transfer process using the Google Admin console is desirable

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Migration tools overview: user led migrations

	Google Workspace Migration for Microsoft Outlook (GWMMO)	Google Workspace Sync for Microsoft Outlook (GWSMO)
Key characteristics	• Calendar, contacts, and PST files	• User installed and managed • Scalable to medium size projects
Who has access?	• All customers	• All customers
When to use?	• User-led migration from a MS Outlook account or PST file.	• Actually its use is not Recommended • Use it only if there's a critical technical dependency on MS Outlook.

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Types of coexistence

01

Short-term internal coexistence

Is necessary during deployment to ensure uninterrupted workflow as people gradually migrate from the legacy system to the new system.

02

Long-term internal coexistence

Is not recommended, but may be required in certain circumstances such as maintaining a legacy application that will not work with Google Workspace.

03

Longer-term external coexistence

Is for situations where the organization works with external bodies that do not use Google Workspace.

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Coexistence duration

Technical team

Duration: Entire deployment timeline

Responsible for:

- Configuring and maintaining both systems
- Addressing technical challenges
- Ensuring a smooth transition for end-users
- Fine-tune configurations before the full cutover

End-users

Duration: approx. 1 month

- This period begins after the Early Adopters (EA) Go Live, when EA users start actively using Google Workspace (GW).
- It ends with the Go Live for Global Go Live (GGL) users, when they are transitioned from the legacy system to GW.

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Coexistence duration

Coexistence for technical team

Phases / Workstreams	Excite (aka Sales)	Enable (aka Deployment) 3 or 4 months			Expand	Embed
		Core IT • 1 month • Max 25 technical users from the IT department	**Early Adopters** • 1 or 2 month(s) • 5% to 10% users representing different roles, departments and locations	**Global Go Live** • 1 month • Everyone else Coexistence for business users		
Governance						
People (Change Mgmt)						
Process (Change Mgmt)						
Technology						
Support						

Go Live for Core IT users

Go Live for EA users

Go Live for GGL users

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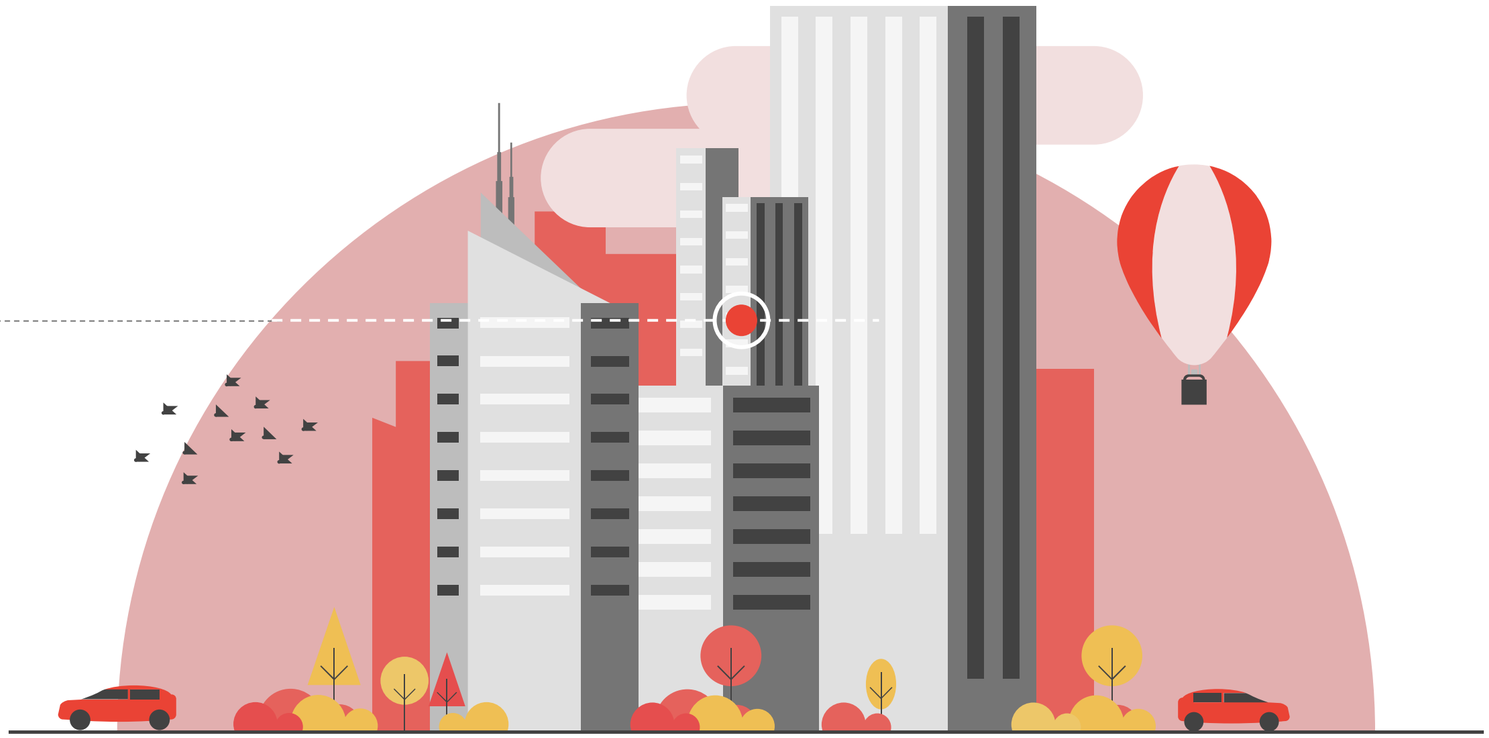
Protecting your organization the right way

Prevention

Detection

Remediation



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